An Ergonomic Vertical Gaming Mouse with Haptic Feedback for Comfortable and Immersive FPS Gameplay

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I. Introduction

Extended use of conventional horizontal mice during first-person shooter (FPS) gameplay has been associated with wrist pronation, musculoskeletal fatigue, and elevated risk of carpal tunnel syndrome [1]. While commercial vertical mice such as the Logitech MX Vertical address wrist posture, none integrate gaming-specific feedback mechanisms suited for competitive FPS use. This project aimed to design, fabricate, and validate a vertically-oriented gaming input device incorporating embedded sensor-actuator electronics and Unity game engine integration, iterating across multiple physical and digital prototypes developed throughout the course.

II. Materials and Methods

A. Prototype Fabrication

The physical prototype was constructed using blue polylactic acid (PLA) filament, 3D printed over approximately 5 hours. The housing consists of a two-piece hollowed handle sculpted across multiple Fusion 360 iterations, sized to accommodate an IR optical mouse sensor (44 mm $\times$ 44 mm) at the base and fit comfortably in an adult hand. The base includes an aperture through which the IR sensor sits flush against the mousing surface. Internal cavity space accommodates all wiring and the Arduino board.

B. Electronics Integration

The electronics system was implemented on an Arduino Uno microcontroller. Three sensors were integrated: a force-sensitive resistor (FSR) to detect applied grip pressure for scroll activation, a potentiometer to determine scroll direction, and a PIR motion sensor to simulate surface movement detection analogous to a traditional optical mouse sensor. Three LED actuators provided output feedback representing the activation states of the primary mouse buttons. Push buttons were used to simulate primary click inputs. Communication was established between the Arduino and Unity. Schematics and bill of materials are provided in Appendix A.

C. Interactive Scene

A Unity-based game scene was developed as the interactive proof-of-concept environment, available at [2]. The scene features use of the controller in an FPS setting. Arduino sensor inputs were mapped to in-game actions as follows: the FSR controls [ACTION], the potentiometer controls [ACTION], and the PIR sensor triggers [ACTION]. LED actuator feedback was triggered upon [EVENT].

D. User Testing

Usability and cognitive load were assessed with $n = 12$ participants using the System Usability Scale (SUS) [3] and the NASA Task Load Index (NASA-TLX) [4]. Participants completed 2 trials with the device while performing the FPS target acquisition task in the Unity scene.

III. Results

The fabricated prototype successfully transmitted sensor data to Unity in real time, allowing the FSR, potentiometer, and PIR inputs to control in-game actions. The Unity scene was operable using the device as the sole input method. Fabrication, CAD files, drawings and assembly details are documented in the Appendix A

A. Usability

SUS scores ranged from [MIN] to [MAX], yielding a mean of $\bar{x} =$ [MEAN] ($SD =$ [SD]). A mean SUS score above 68 is considered above average [3]. [BRIEF INTERPRETATION — e.g., "indicating participants found the device usable with a minor learning curve."]

B. Cognitive Load

NASA-TLX results indicated an overall workload score of [SCORE] out of 100. The highest contributing subscale was [SUBSCALE] ($M =$ [VALUE]), suggesting [INTERPRETATION]. Full results are presented in Appendix A.

IV. Takeaways and Recommendations

The iterative design process from the initial paper prototype through to the final 3D-printed handle demonstrated that physical constraints influenced component placement decisions. The IR sensor's 44 mm $\times$ 44 mm placement required two handle redesigns before a layout was found that accommodated it at the base without compromising grip ergonomics. The integration of the components proved more challenging than anticipated due to space to work with, which was resolved by iteration.

Future versions will consider replacing the PIR motion sensor with a dedicated optical flow sensor for more accurate cursor simulation, integrating tactile buttons directly into the printed shell rather than mounting them separately, and conducting extended wear trials to better assess long-term benefit over typical mouse use. User feedback indicated that [AJ: fill in after testing — common themes].

References

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Appendix

The prototype was fabricated using blue PLA filament via FDM 3D printing. Total print time was approximately 5 hours. The design progressed through several major revisions before the final print: an initial paper prototype was built to evaluate grip comfort, followed by a Blender model, then two Fusion 360 iterations (see Appendix A for full CAD progression). The primary fabrication challenge was accommodating the IR mouse sensor (44 mm $\times$ 44 mm) in the base while maintaining the handle profile — resolved in the final iteration by enlarging the base footprint and fully hollowing the interior.

Total fabrication cost: approximately \$10 CAD (\$9 IR mouse sensor, $\$\sim 1PLAfilament).Totaltimeinvested : 15hours(10hoursmodelingacrossalliterations, 5hoursprinting)$

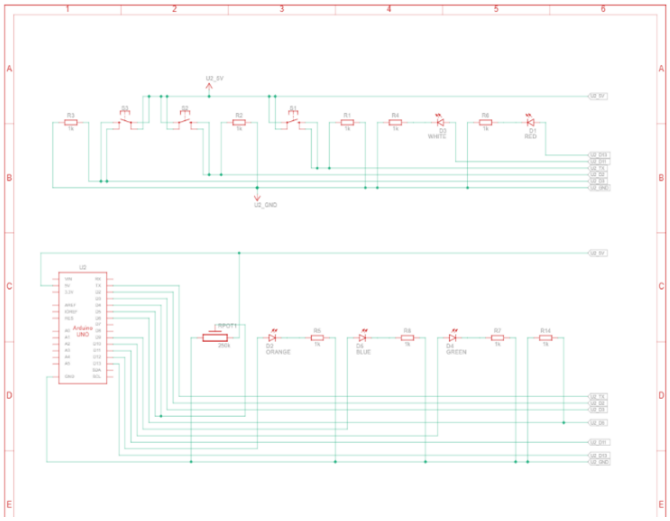

Fig. 1. Electronics schematic for the vertical mouse prototype, showing Arduino Uno connected to FSR force sensor, potentiometer, PIR motion sensor, and three LED actuators.

As shown in Fig. **??**, the Arduino Uno reads analog input from the FSR (pins 4 and 6) and potentiometer (pins 5 and 6), and digital input from the PIR sensor and three push buttons (pins 1, 2, 3). Three LEDs on output pins serve as actuators indicating button activation state. [ALEX: update pin descriptions if anything changed from A1 to the final physical build.]

TABLE I
BILL OF MATERIALS

Component	Qty	Cost (CAD)	Source
Arduino Uno	1	[X]	[Store]
FSR Force Sensor	1	[X]	[Store]
Potentiometer	1	[X]	[Store]
PIR Motion Sensor	1	[X]	[Store]
LEDs (assorted)	3	[X]	[Store]
Push buttons	3	[X]	[Store]
IR Mouse Sensor	1	\$9.00	[Store]
Blue PLA Filament	—	∼\$1.00	[Store]
Breadboard	1	[X]	[Store]
Resistors (assorted)	[X]	[X]	[Store]
Jumper wires	[X]	[X]	[Store]
Total		**∼\$10.00**	

```
// Vertical Mouse Arduino Code
// Pins 1,2,3: push button inputs (click simulatio
// Pins 11,12,13: LED outputs (button activation s
// Pins 4,6: FSR force sensor (scroll activation t
// Pins 5,6: Potentiometer (scroll direction)
// Pins 9,10: LED outputs (scroll direction feedba

void setup() {
  pinMode(1, INPUT);    // Button 1 (left click)
  pinMode(11, OUTPUT); // LED 1
  pinMode(2, INPUT);    // Button 2 (right click)
  pinMode(12, OUTPUT); // LED 2
  pinMode(3, INPUT);    // Button 3 (side button / 
```

```
  pinMode(13, OUTPUT); // LED 3
  pinMode(4, INPUT);   // FSR low threshold
  pinMode(6, INPUT);   // FSR high threshold
  pinMode(10, OUTPUT); // Scroll up LED
  pinMode(5, INPUT);   // Potentiometer
  pinMode(9, OUTPUT);  // Scroll down LED
}

void loop() {
  // Button 1 - Left click
  if (digitalRead(1) == HIGH) {
    digitalWrite(11, HIGH);
  } else {
    digitalWrite(11, LOW);
  }

  // Button 2 - Right click
  if (digitalRead(2) == HIGH) {
    digitalWrite(12, HIGH);
  } else {
    digitalWrite(12, LOW);
  }

  // Button 3 - Side button
  if (digitalRead(3) == HIGH) {
    digitalWrite(13, HIGH);
  } else {
    digitalWrite(13, LOW);
  }

  // FSR + Potentiometer - Scroll up
  if (digitalRead(4) <= 0.1 && digitalRead(6
    digitalWrite(10, HIGH);
  } else {
    digitalWrite(10, LOW);
  }

  // FSR + Potentiometer - Scroll down
  if (digitalRead(5) >= 1 && digitalRead(6)
    digitalWrite(9, HIGH);
  } else {
    digitalWrite(9, LOW);
  }

  delay(10); // Stabilize loop timing
}
```

The interactive scene is a Unity-based game developed as part of the GDW Year 3 project, available at [2].

The prototype evolved through the following documented stages:

1) **Initial Unity concept model** — A rough 3D visualization built from Unity primitive shapes. Featured a flat baseboard, a screen at the top, two side buttons, a scroll wheel, and one front button. Used to establish initial spatial layout.

TABLE II
INPUT MAPPING: ARDUINO SENSORS TO UNITY ACTIONS

Physical Input	Sensor	Unity Action
Grip squeeze	FSR Force Sensor	[ACTION]
Dial rotation	Potentiometer	[ACTION]
Motion detection	PIR Sensor	[ACTION]
Button 1 press	Push button	[ACTION]
Button 2 press	Push button	[ACTION]
Button 3 press	Push button	[ACTION]

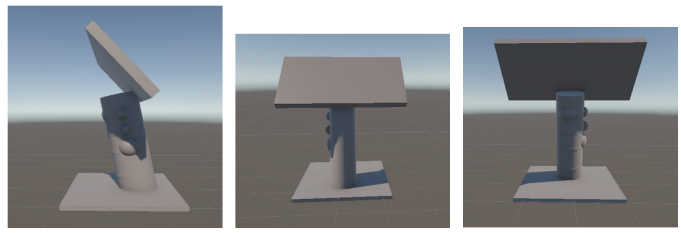

Fig. 2. Initial Unity concept model — three angles showing the flat baseboard, top-mounted screen, side scroll wheel, and button layout.

2) **Physical paper prototype** — A cardboard/paper model with a thicker baseboard and more vertical screen. No buttons at this stage. Used to evaluate grip comfort and overall form factor.

Fig. 3. Physical paper prototype showing thicker baseboard and more vertical screen orientation.

3) **Blender model** — Translated the paper prototype into 3D. Reduced screen size, refined baseboard geometry, established button placement based on paper prototype feedback.
4) **First Fusion 360 model** — Pivoted from the baseboard-and-screen concept to a pistol-grip style handle. First version without the secondary display component.

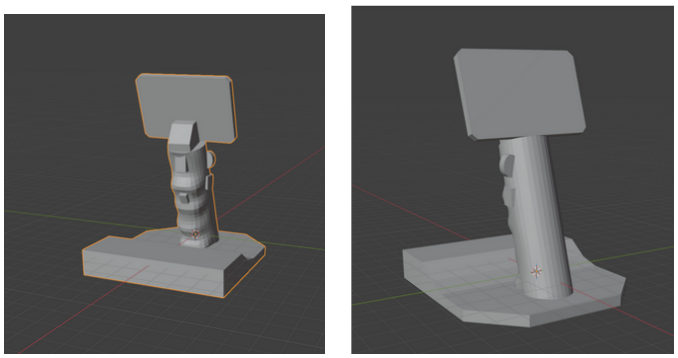

Fig. 4. Blender model — two angles showing the refined baseboard, reduced screen size, and button placement.

Fig. 5. First Fusion 360 model — initial pistol-grip handle geometry, first iteration without baseboard or screen.

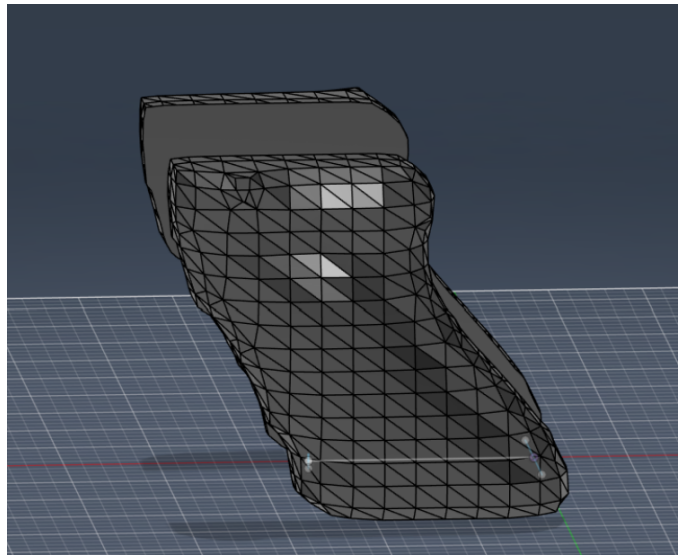

Fig. 6. Second Fusion 360 model — more defined grip profile approaching the final handle shape.

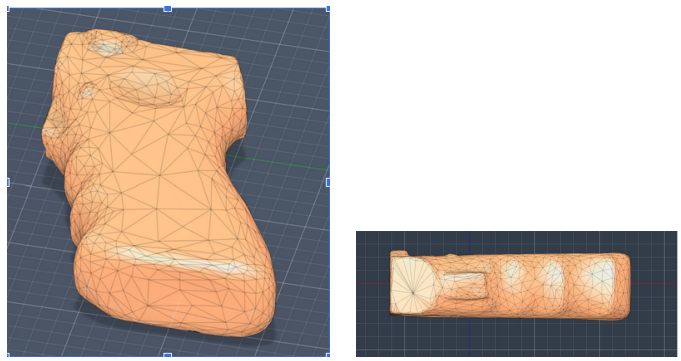

Fig. 7. Intermediate sculpt iterations refining the handle ergonomics and internal cavity geometry.

5) **Second Fusion 360 model** — Refined handle geometry with a more defined grip profile, closely matching the shape of the final product.
6) **Final Fusion 360 model** — Hollowed interior to house electronics and wiring. Base enlarged to accommodate the 44 mm $\times$ 44 mm IR mouse sensor. Shell split into two halves for assembly access. Sized to fit most adult hands.
7) **3D print** — Final model printed in blue PLA, approximately 5 hours print time.
8) **Electronics integration** — Arduino Uno assembled with FSR, potentiometer, PIR sensor, and three LED actuators. [JOSEPH/ALEX: add any notes on challenges here]
9) **Unity scene finalized** — GDW game confirmed as interactive demo environment. Input mapping implemented.
10) **User testing conducted** — 12 participants assessed with SUS and NASA-TLX.
11) **Final submission** — [DATE]

Team Roles and Responsibilities:

- AJ Bwejesa: Documentation, report writing, user testing coordination
- Alex Bean: Electronics, Arduino code, TinkerCAD simulation, CAD design
- Joseph Chanine: Physical fabrication, hardware engineering, 3D modeling

The prototype was modeled across multiple iterations in Fusion 360 (see Appendix A for full progression). The final design consists of two mirrored shell halves, hollowed to house the electronics assembly, with a base aperture for the IR sensor.

Insert the following:

- Part blueprints with dimensions (ISO standard — export from Fusion 360 drawing mode)
- Assembly view showing both shell halves with electronics positioned inside
- Exploded view showing how the two halves separate and how the IR sensor, Arduino, and wiring are seated
- Electronic assembly simulation video: [LINKIFEXISTS]

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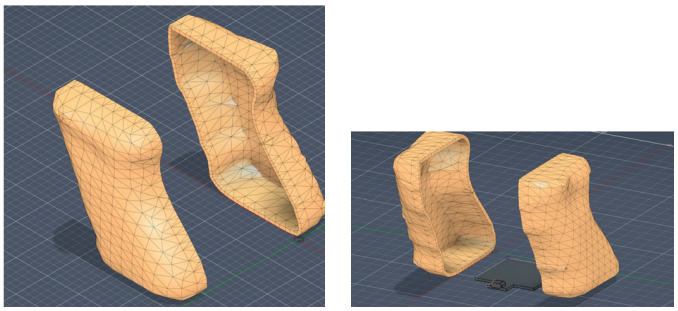

Fig. 8. Final Fusion 360 model — hollowed two-piece handle with base aperture for IR sensor, shown assembled and exploded.

TABLE III
SUS SCORES PER PARTICIPANT

Participant	SUS Score (/100)
P1	[SCORE]
P2	[SCORE]
P3	[SCORE]
P4	[SCORE]
P5	[SCORE]
P6	[SCORE]
P7	[SCORE]
P8	[SCORE]
P9	[SCORE]
P10	[SCORE]
P11	[SCORE]
P12	[SCORE]
Mean	**[MEAN]**

We declare that all team members contributed meaningfully to this project. The specific contributions are as follows:

AJ Bwejesa: Planning, report writing and structuring, user testing coordination and analysis.

Alex Bean: Planning, electronics assembly and wiring, Arduino code development, TinkerCAD simulation, CAD modeling.

Joseph Chanine: Planning, physical prototype fabrication, hardware engineering, 3D modeling and Fusion 360 iterations, 3D printing.

AI Usage Disclosure: All technical content, descriptions, and analysis were written and verified by the team.

TABLE IV
NASA-TLX SCORES PER PARTICIPANT (0–100 SCALE)

P#	Mental	Physical	Temporal	Perf.	Effort	Frustration	Overall
P1							
P2							
P3							
P4							
P5							
P6							
P7							
P8							
P9							
P10							
P11							
P12							
Mean							